화학수학

열 번째 수업

미분방정식

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Terminology

- Order of DE (계수): order of the highest derivative

- Linear (선형) vs. nonlinear (비선형)
  - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

  - Nonlinear: otherwise

$$(1-y)\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = e^x, \qquad \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \sin y = 0, \qquad \cdots$$

Linear Differential Equations

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions
(일반해와 특수해)

- Solution: any function that reduces the DE to an identity when substituted into the DE

For a DE of order n,

- General solution: contains n arbitrary constants
- Particular solution: contains no arbitrary constants
  - Particular solutions can be generated from the general solution

Not in Mortimer

Additional Conditions

- For a DE of order n, if n additional conditions are given, one can determine the arbitrary constants of the general solution

- Initial conditions (IC, 초기 조건): values of the function and its first $(n-1)$ derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Solving a Differential Equation

1. Obtain the general solution: hardest part!
2. If initial/boundary conditions are given, substitute the conditions to determine the coefficients

- No universal method to solve all DEs
- For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y)\frac{\mathrm{d}y}{\mathrm{d}x} = f(x)$$

This DE can be manipulated into

$$g(y)\,\mathrm{d}y = f(x)\,\mathrm{d}x$$

$$\int g(y)\,\mathrm{d}y = \int f(x)\,\mathrm{d}x + C$$

This equation provides the general solution

Exact Differential Equations

- Pfaffian form:

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

If the differential equation is exact,

there is a function $f(x,y)$ such that

$$\mathrm{d}f = M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

which implies that $f(x,y) = k$ (constant)

How to Find $\boldsymbol{f(x, y)}$

$$\mathrm{d}f = M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C \mathrm{d}f$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1). Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1):

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0)\,\mathrm{d}x + \int_{y_0}^{y_1} N(x_1, y)\,\mathrm{d}y$$

This gives the relationship between x and y.

Integrating Factors

- Pfaffian form:

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

If the differential equation is inexact, but we know the integration factor $g(x,y)$ that makes

$$g(x,y)M(x,y)\,\mathrm{d}x + g(x,y)N(x,y)\,\mathrm{d}y = 0$$

an exact differential equation, we can solve this DE.

Not in Mortimer

Linear Equations

$$a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)\mathrm{d}x} \int e^{\int P(x)\mathrm{d}x} f(x)\,\mathrm{d}x + Ce^{-\int P(x)\mathrm{d}x}$$

Not in Mortimer

Higher-Order ODEs: Linear Case

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
  - Homogeneous (§ 12.2)
  - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs

Homogeneous Linear Equations

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \cdots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \cdots, c_k$.

Not in Mortimer

Reduction of Order

$$a_2(x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x}+a_0(x)y=0$$

If a solution y_1 is known,

we can obtain the other solution y_2

$$y_2(x)=y_1\int\frac{1}{y_1^2}e^{-\int P\mathrm{d}x}\,\mathrm{d}x$$

오늘의 수업

- Homogeneous linear ODEs with constant coefficients
  - Solving strategy
  - Application: harmonic oscillator
  - Application: particle in a box

- Heterogeneous linear ODEs with constant coefficients
  - Solving strategy
  - Application: harmonic oscillator

Linear ODEs with Constant Coefficients

- Homogeneous

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+b\frac{\mathrm{d}y}{\mathrm{d}x}+cy=0$$

- Inhomogeneous

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+b\frac{\mathrm{d}y}{\mathrm{d}x}+cy=g(x)$$

Homogeneous Case

$$a\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Roots of Characteristic Equation

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1. Two roots are real and distinct ($b^2 - 4ac > 0$)
2. Two roots are real and equal ($b^2 - 4ac = 0$)
3. Two roots are complex numbers ($b^2 - 4ac < 0$)

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

Example 12.2

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0.$$

The characteristic equation is

$$\lambda^2 + \lambda - 2 = 0$$

whose solutions are

$$\lambda_1 = 1,\ \lambda_2 = -2$$

Hence, the general solution is

$$y = c_1 e^{x} + c_2 e^{-2x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P \mathrm{d}x}}{y_1^2} \mathrm{d}x = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} \mathrm{d}x = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 10\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 0.$$

The characteristic equation is

$$\lambda^2 - 10\lambda + 25 = 0$$

whose solution is

$$\lambda = 5$$

Hence, the general solution is

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

(3) Conjugate Complex Roots

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

Since $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$,

$$y = c_1 e^{\alpha x}(\cos\beta x + i\sin\beta x) + c_2 e^{\alpha x}(\cos\beta x - i\sin\beta x)$$

$$= e^{\alpha x}[\underbrace{(c_1 + c_2)}_{c_1'}\cos\beta x + \underbrace{i(c_1 - c_2)}_{c_2'}\sin\beta x]$$

$$y = e^{\alpha x}(c_1' \cos\beta x + c_2' \sin\beta x)$$

Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 7y = 0.$$

The characteristic equation is

$$\lambda^2 + 4\lambda + 7 = 0$$

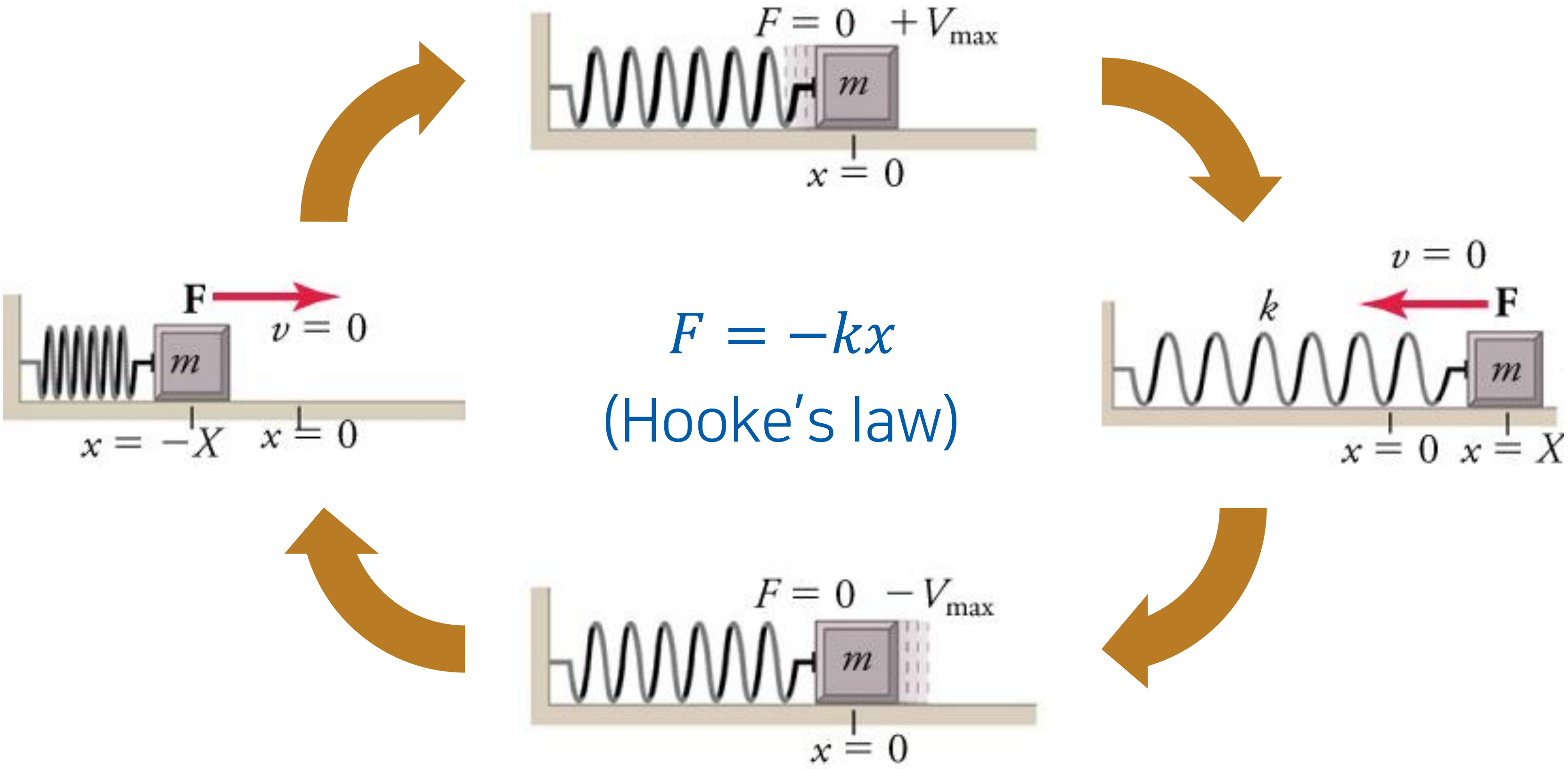
whose solutions are

$$\lambda_1 = -2 + i\sqrt{3},\ \lambda_2 = -2 - i\sqrt{3}$$

Hence, the general solution is

$$y = e^{-2x}\left[c_1 \cos\left(\sqrt{3}x\right) + c_2 \sin\left(\sqrt{3}x\right)\right]$$

Application: Harmonic Oscillator (조화진동자)



$$F = -kx$$

(Hooke's law)

(Image: https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion)

Equation of Motion (운동방정식)

- Newton's second law

$$F = ma = m\frac{\mathrm{d}v}{\mathrm{d}t} = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2}$$

- Equation of motion

$$F = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -kx$$

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

Solving Equation of Motion

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m} x = 0$$

The characteristic equation is

$$\lambda^2 + \frac{k}{m} = 0$$

whose solutions are

$$\lambda_1 = i\sqrt{k/m}, \qquad \lambda_2 = -i\sqrt{k/m}$$

Solving Equation of Motion

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m} x = 0$$

$$\lambda_1 = i\sqrt{k/m}, \qquad \lambda_2 = -i\sqrt{k/m}$$

Define $\omega = \sqrt{k/m}$; the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Assume that we know the conditions at $t = 0$:

$$x(0) = 0, \qquad v(0) = \left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} = v_0$$

From the first condition,

$$x(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0$$

$$x(t) = c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_2 \sin \omega t$$

$$v(0) = \left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{t=0} = v_0$$

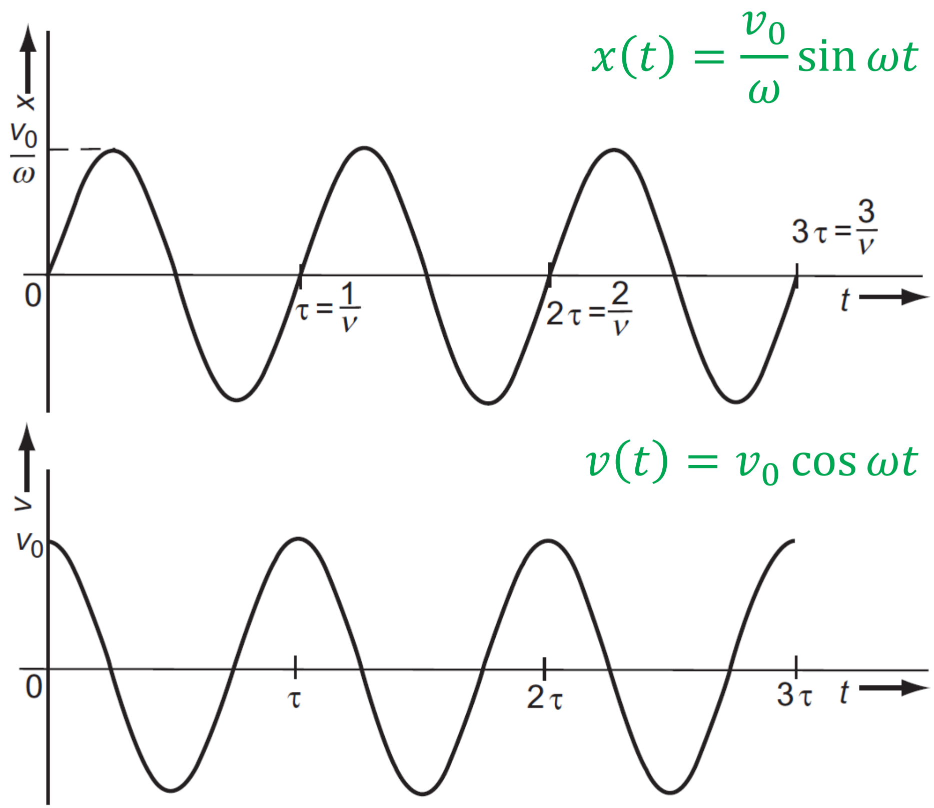
From the second condition,

$$v(0) = c_2 \omega \cos 0 = c_2 \omega = v_0$$

Hence, the solution is

$$x(t) = \frac{v_0}{\omega} \sin \omega t$$

Uniform Harmonic Motion

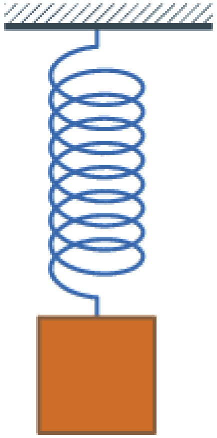


(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.1)

Damped Harmonic Oscillator

(감쇠 조화진동자)

- Damping (감쇠): reduction of oscillation
- Application of friction force (마찰력)
  - proportional to the velocity
  - in the opposite direction



$$\vec{F}_f = -\zeta\vec{v} = -\zeta\frac{\mathrm{d}\vec{r}}{\mathrm{d}t}$$

friction constant
(마찰 상수)

(Image: https://commons.wikimedia.org/wiki/File:Damped_spring.gif)

1-Dimensional Equation of Motion

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx - \zeta\frac{\mathrm{d}x}{\mathrm{d}t}$$

- Standard form

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\zeta}{m}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{k}{m}x = 0$$

- Characteristic equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

Roots of Characteristic Equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

$$D = \left(\frac{\zeta}{m}\right)^2 - \frac{4k}{m}$$

1. Distinct real roots ($D > 0$)
2. Repeated real roots ($D = 0$)
3. Conjugate complex roots ($D < 0$)

(1) $\boldsymbol{D > 0}$: Overdamping (과감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \qquad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

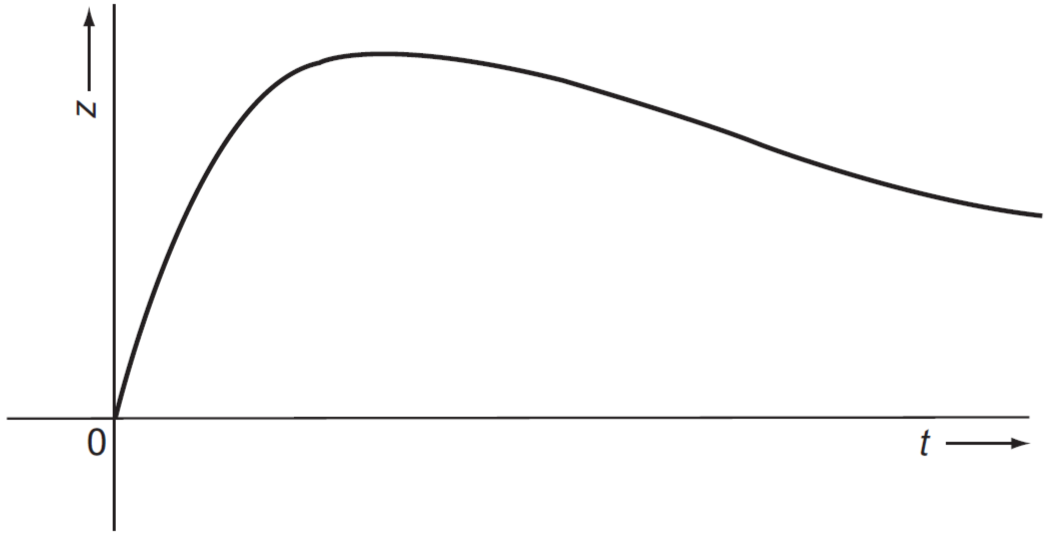
$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

- Note that $\lambda_1, \lambda_2 < 0$ (Example 12.6) → decay!

- If the initial condition is $x(0) = 0$, $c_1 = -c_2$

(1) $D > 0$: Overdamping (과감쇠)

$$x(t) = c(e^{\lambda_1 t} - e^{\lambda_2 t})$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.2)

(2) $\boldsymbol{D = 0}$: Critical Damping (임계감쇠)

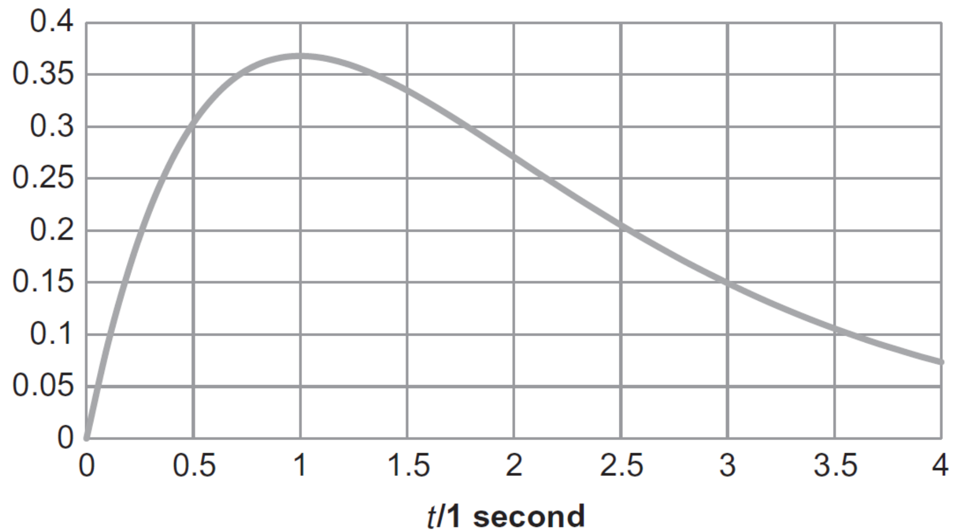
$$\lambda_1 = -\frac{\zeta}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$$

- Note that $\lambda_1 < 0$ → decay!
- If the initial condition is $x(0) = 0$, $c_1 = 0$

(2) $D = 0$: Critical Damping (임계감쇠)

$$x(t) = c_2 t e^{\lambda_1 t}$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.4)

(3) $\boldsymbol{D} < \mathbf{0}$: Underdamping (미급감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4k}}{2m}, \qquad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4k}}{2m}$$

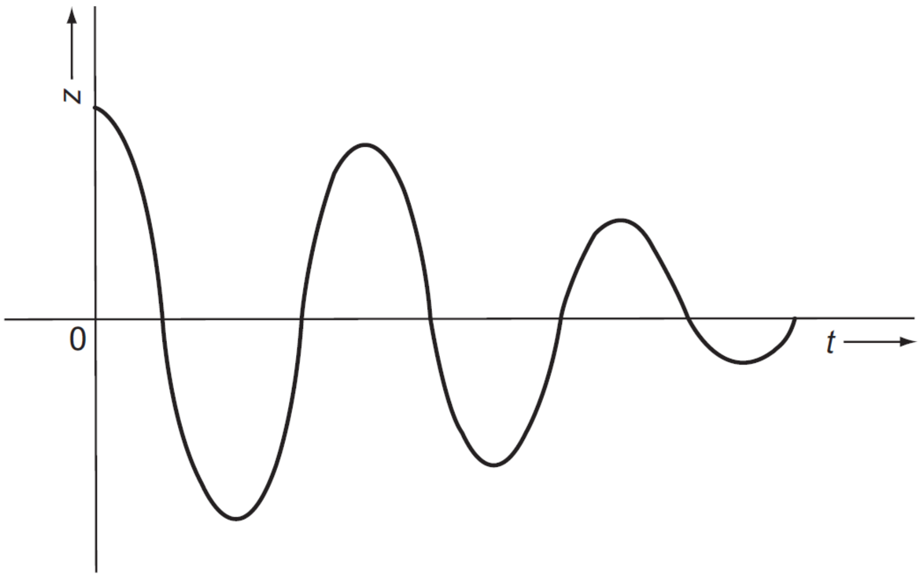
Let $\omega = \sqrt{(k/m) - (\zeta/2m)^2}$;

$$\lambda_1 = -\frac{\zeta}{2m} + i\omega, \qquad \lambda_2 = -\frac{\zeta}{2m} - i\omega$$

$$x(t) = e^{-\zeta t/2m}(c_1 \cos \omega t + c_2 \sin \omega t)$$

(3) $D < 0$: Underdamping (미급감쇠)

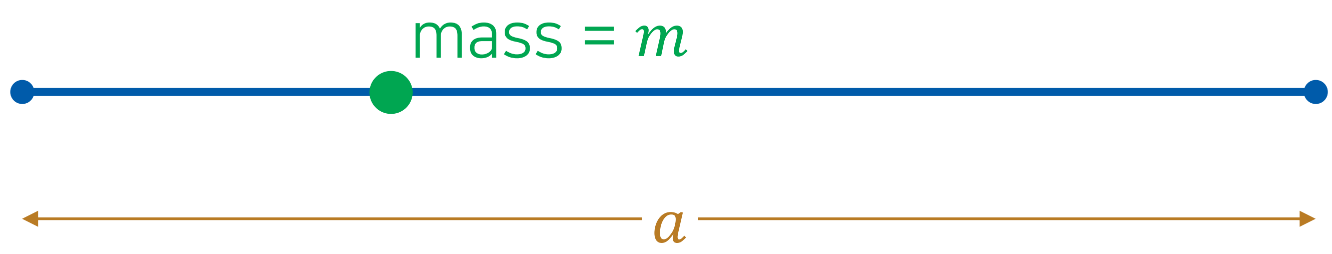
$$x(t) = ce^{-\zeta t/2m} \cos \omega t$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.3)

Not in Mortimer

Application: Particle in a Box



$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

Boundary conditions: $\psi(0) = \psi(a) = 0$

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Solving the Differential Equation

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

- Characteristic equation: $\lambda^2 + \frac{2mE}{\hbar^2} = 0$

Since $E > 0$, the characteristic equation has two conjugate complex roots:

$$\lambda_1 = \frac{i\sqrt{2mE}}{\hbar}, \qquad \lambda_2 = -\frac{i\sqrt{2mE}}{\hbar}$$

General Solution

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2}+\frac{2mE}{\hbar^2}\psi(x)=0 \quad (E>0)$$

$$\lambda_1=\frac{i\sqrt{2mE}}{\hbar}, \qquad \lambda_2=-\frac{i\sqrt{2mE}}{\hbar}$$

Hence, the general solution is

$$\psi(x)=c_1\cos\left(\frac{\sqrt{2mE}}{\hbar}x\right)+c_2\sin\left(\frac{\sqrt{2mE}}{\hbar}x\right)$$

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Boundary Conditions

$$\psi(x) = c_1 \cos\left(\frac{\sqrt{2mE}}{\hbar}x\right) + c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar}x\right)$$

- Boundary conditions: $\psi(0) = \psi(a) = 0$

$$\psi(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0$$

$$\psi(a) = c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar}a\right) = 0$$

To avoid $c_2 = 0$, we obtain $\frac{\sqrt{2mE}}{\hbar}a = n\pi$ (n: integer)

Final Solution

$$\frac{\sqrt{2mE}}{\hbar}a = n\pi$$

$$E = \frac{\hbar^2\pi^2}{2ma^2}n^2$$

The final solution is

$$\psi(x) = c_2 \sin\left(\frac{n\pi x}{a}\right)$$

(One can use normalization to determine c_2.)

Inhomogeneous Linear ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = \underline{g(x)}$$

inhomogeneous term
(비동차항)

- Complementary equation (보상방정식)

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

Solving Inhomogeneous ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+b\frac{\mathrm{d}y}{\mathrm{d}x}+cy=g(x)$$

1. Solve the complementary equation →
   complementary function (보상함수) y_{c}
2. Find a particular solution y_{p} that satisfies the DE
3. The general solution is $y=y_{\mathrm{c}}+y_{\mathrm{p}}$

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Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 16x.$$

A particular solution to this equation is $y = 4x + 5$.

The complementary equation is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

and the complementary function is

$$y_{\mathrm{c}}(x) = c_1 e^x + c_2 e^{4x}$$

Hence, the general solution is

$$y = c_1 e^x + c_2 e^{4x} + 4x + 5$$

Method of Undetermined Coefficients

(미정계수법)

How can we obtain a particular solution y_{p}?

Guess the form of y_{p} with undetermined coefficients!

Trial Solutions

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 12.1)

Not in Mortimer

Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 2x^2 - 3x + 6.$$

The complementary equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

and the complementary function is

$$y_\mathrm{c}(x) = c_1e^{-2x} + c_2xe^{-2x}$$

Not in Mortimer

Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 2x^2 - 3x + 6.$$

$$y_\mathrm{c}(x) = c_1 e^{-2x} + c_2 x e^{-2x}$$

Now, consider a trial solution $y_\mathrm{p} = A_0 + A_1 x + A_2 x^2$, whose derivatives are

$$\frac{\mathrm{d}y_\mathrm{p}}{\mathrm{d}x} = A_1 + 2A_2 x, \qquad \frac{\mathrm{d}^2 y_\mathrm{p}}{\mathrm{d}x^2} = 2A_2$$

Substitution leads to

$$4A_2 x^2 + (8A_2 + 4A_1)x + (4A_0 + 4A_1 + 2A_2) = 2x^2 - 3x + 6$$

Comparison gives $A_2 = 1/2\,, A_1 = -7/4\,, A_0 = 3$.

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Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 2x^2 - 3x + 6.$$

$$y_\mathrm{c}(x) = c_1e^{-2x} + c_2xe^{-2x}$$

$$y_\mathrm{p}(x) = \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

The general solution is

$$y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

Not in Mortimer

Glitch in the Method

A trial solution y_p can be

a solution of the complementary equation.

In this case, try $x^n y_\mathrm{p}$ instead!

(n: smallest integer that resolves the problem)

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Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 8e^x.$$

The complementary function is

$$y_\mathrm{c}(x) = c_1e^x + c_2e^{4x}$$

According to Table 12.1, the trial solution for $8e^x$ is Ae^x.

However, substitution gives

$$Ae^x - 5Ae^x + 4Ae^x = 0 = 8e^x$$

The trial solution is a wrong guess.

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Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 8e^x.$$

$$y_\mathrm{c}(x) = c_1 e^x + c_2 e^{4x}$$

Instead of Ae^x, we can try $y_\mathrm{p} = Axe^x$.

$$\frac{\mathrm{d}y_\mathrm{p}}{\mathrm{d}x} = Ae^x + Axe^x, \qquad \frac{\mathrm{d}^2 y_\mathrm{p}}{\mathrm{d}x^2} = 2Ae^x + Axe^x$$

Hence,

$$\frac{\mathrm{d}^2 y_\mathrm{p}}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y_\mathrm{p}}{\mathrm{d}x} + 4y_\mathrm{p} = -3Ae^x$$

and $A = -8/3$.

Example

Find the general solution to the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 5\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 8e^x.$$

$$y_\mathrm{c}(x) = c_1e^x + c_2e^{4x}$$

$$y_\mathrm{p}(x) = -\frac{8}{3}xe^x$$

Therefore, the general solution is

$$y(x) = c_1e^x + c_2e^{4x} - \frac{8}{3}xe^x$$

Application: Forced Harmonic Oscillator

$$F = m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -kx + \underbrace{F(t)}_{\text{external force}}$$

- Standard form

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m}x = \frac{F(t)}{m}$$

- Complementary equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

$$\rightarrow x_\mathrm{c}(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

The equation of motion is

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m} x = \frac{F(t)}{m} = \frac{F_0}{m} \sin \alpha t$$

The trial function for $\sin \alpha t$ is $A \sin \alpha t + B \cos \alpha t$. Substitution gives

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m}(A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m}(A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Rearranging the equation,

$$\left(-\alpha^2 + \frac{k}{m}\right) A \sin \alpha t + \left(-\alpha^2 + \frac{k}{m}\right) B \cos \alpha t = \frac{F_0}{m} \sin \alpha t$$

Hence, $B = 0$ and $A = \frac{F_0/m}{-\alpha^2 + k/m} = \frac{F_0}{k - m\alpha^2}$.

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

$$x_p = \frac{F_0}{k - m\alpha^2} \sin \alpha t$$

As we know $x_c(t) = c_1 \cos \omega t + c_2 \sin \omega t$, the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{k - m\alpha^2} \sin \alpha t$$